

ADPMod

A harmonic ring modulator with five carrier waves, tape-style drive and a tone filter.



Overview

ADPMod is a ring modulator: it multiplies your signal by a carrier wave. At audio-rate carrier frequencies this turns notes into clangorous, bell-like and metallic tones that no longer follow the original pitch; at very low carrier frequencies it becomes a pulsing, tremolo-like effect. The carrier can be a sine, triangle, square, saw or noise, and HARMONICS adds the carrier's octave on top for a richer, more complex result.

After the modulation, the effected (wet) signal goes through a tape-style DRIVE stage and a TONE low-pass filter before it is blended with the dry signal. A slow random 'chaos drift' moves the carrier frequency, the character of the drive and the tone filter, so the effect never sits completely still. A low cut on the blend keeps the bass clean if you want it to.

Reach for it for

- Metallic, bell-like and robotic vocals
- Clangorous textures from synths, guitars and drums
- Slow, pulsing tremolo with low carrier frequencies
- Wide, shimmering stereo modulation (DUAL)
- Adding grit and movement to sound design

Controls

Modes (header)

AUDIO	One carrier for both sides, so the result stays mono-compatible. The default.
SUB	The same ring modulator. Use it with low FREQ settings, where the carrier is below the audible range and the effect becomes a pulse or tremolo.
DUAL	The right side's carrier runs very slightly detuned (0.5%) from the left, so the two sides drift against each other for a wider stereo image.
OFF	The ring modulator is switched off: the audio passes through the input and output stages only.

The carrier

CARRIER WAVEFORM display	Shows the carrier wave, moving with it.
SIN / TRI / SQR / SAW / NOISE	The carrier wave. SIN gives the purest ring-mod tones; TRI is a little brighter; SQR and SAW are harsh and buzzy, with many more side tones; NOISE turns the signal into a gritty, broken texture.
FREQ	The carrier frequency, 0.5 Hz to 2 kHz (220 Hz by default). Low settings give a pulse or tremolo; audio-rate settings give metallic, bell-like tones.
DRIVE	A tape-style saturation stage on the modulated (wet) signal, 0 to 10 (0 by default). It warms the sound at low settings and adds thickness and grit as you turn it up.
DEPTH	How much the carrier modulates the signal, 0 to 100% (100% by default). 100% is full ring modulation; lower settings keep more of the unmodulated signal in the wet sound.

HARMONICS	Adds the carrier's octave (twice its frequency) to the carrier, 0 to 100% (50% by default), for a denser, more complex set of tones.
TONE	A low-pass filter on the wet signal, 20 Hz to 18 kHz (1 kHz by default). Turn it down to tame harsh side tones; turn it up for a brighter, more metallic effect.

Drift

CHAOS DRIFT: a slow random wander of the carrier frequency, the character of the drive and the TONE filter.

DRIFT	How much drift, 0 to 100%. 50% (the default) is the standard amount, 100% is twice that, 0 switches it off. More drift also makes the wander a little faster.
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IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right and is lit while it is on. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light glows brighter while it is catching peaks.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

Master

MIX	Dry / wet balance, 0 to 100% (50% by default).
LOW CUT	Keeps the lows dry: below this frequency the effect is not blended in, so the bass stays clean and solid. From off (fully left, 20 Hz) to 500 Hz.

OUTPUT	Output level, -24 to +12 dB.
LEVEL meters	Output level, left and right.

The header

PRESETS	Opens the list of factory presets (a click on the display does the same). Choosing one sets the mode, carrier, drive, depth, harmonics, tone and mix; drift, LOW CUT, output and the IN / OUT section are left as they are.
◀ / ▶	Step to the previous or next preset.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Vocal Ring (Classic)	AUDIO mode, 440 Hz sine, light drive, 30% harmonics, tone at 2.5 kHz, 70% mix.
Sub Tremolo	SUB mode, 6.5 Hz triangle at 80% depth with some drive: a throbbing tremolo, 50% mix.
Metallic Chaos	DUAL mode, 180 Hz square, heavy drive and 85% harmonics, bright tone, 85% mix.
Subtle Saturation Mod	AUDIO mode, 220 Hz sine at 50% depth, no harmonics, open tone, 40% mix.

Tips

- For tuned, musical results, set **FREQ** to a note that fits the key of the song (for example 220 Hz or 440 Hz for A).

- Sweep FREQ slowly with automation for dramatic, sci-fi movement.
- On a bass or a full mix, raise LOW CUT to 150-300 Hz so the low end stays dry and solid.
- If the effect is too harsh, lower TONE or HARMONICS before turning down MIX.

File: gfx-adpmod.jsfx